# How to Achieve High Availability

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## How to Achieve High Availability?

Fundamentally, high availability is about continuing to operate without failures. There are two types of failures:

Expected failures can include:

- Hardware failures. Despite the best efforts, hardware components can and do fail unexpectedly. This can be due to various reasons such as manufacturing defects, wear and tear, or environmental conditions.
- Resource Exhaustion: If your server's disk space or memory may get filled up at a particular load, that's a predictable failure. These can be mitigated by monitoring resource usage and scaling or cleaning up resources as necessary.

Unexpected failures can include

- A misbehaving client. For example, a hacker who tries to exhaust our resources by sending a flood of requests to our API endpoint. A rate limiter is typically deployed in such cases to limit the number of calls a client can make.
- A failure in our service’s dependency: Our system has a dependency on another system if it needs to interact with an external system. A failure of the external system should not cause a failure in our system. For example, a hotel booking service may have to interact with 3rd party partner systems to complete a hotel booking. Our system should be able to handle failures in the 3rd party system.

## Handling Expected Systems Failures

Expected failures can be mitigated by establishing proper redundancy. This can be achieved by setting up your service across multiple regions, load balancing your stateless servers, and setting up automatic failover.

## Redundancy

Redundancy is a critical part of achieving high availability in a system. It's all about having backup resources, like servers or databases, that are ready to take over if the primary resources fail. When dealing with cloud environments, there are two key concepts related to redundancy: availability zones and regions.

What is an availability zone?

An Availability Zone (AZ) is a distinct location within a cloud provider's region, insulated from failures in other availability zones. Each availability zone runs on its own physically separate, independent infrastructure, and is engineered to be highly reliable. In the event of a failure, services can be transitioned to a different zone within the same region.

A Region is a geographical area that consists of multiple, isolated availability zones. Deploying applications across multiple regions provides greater fault tolerance and latency reduction as each region operates independently. This means a problem in one region doesn't affect another region.

To achieve high availability, services are often deployed across multiple availability zones within a region, which protects from single points of failure. For an even higher level of redundancy, services can be deployed across multiple regions. This can protect against larger-scale issues like natural disasters that might affect an entire region.

## Load Balancing

Load balancing refers to the distribution of network traffic across multiple servers. It prevents any single server from becoming overworked (a bottleneck), which could lead to a system failure.

In cloud environments, load balancers can be deployed within a single availability zone or across multiple zones and regions, ensuring traffic is evenly distributed and providing an automatic failover mechanism. If a server or entire zone fails, the load balancer can redirect traffic to the remaining operational servers or zones.

## Data Replication

Data replication involves maintaining copies of your data on different databases or database servers. If your primary database fails, one of the replicated databases can take over, ensuring your system continues to have access to its data.

In a cloud environment, you can configure data replication across multiple servers within an availability zone, across multiple zones, or even across regions, providing an additional level of redundancy and high availability.

## Health Monitoring and Auto-Recovery Systems

Health monitoring and recovery systems automatically check the health of servers and other system components. If they detect a failure, they can automatically initiate recovery procedures, such as restarting a service or triggering a failover to a backup resource.

Cloud providers often offer services that monitor the health of your applications and automatically recover failed instances. For example, Amazon EC2 Auto Recovery automatically recovers instances when a system impairment is detected.

In practice, we often use services or products that are a combination of the above. Let’s take a look at the tech stacks commonly used for high availability.